```
strong><br />
A can of tuna.</li>
</ul>
</body></html>
```

CommonMark-generated markup starts from *<h2>* and ends in *
*. The rest is the HTML template. Figure 2 shows what the HTML looks like in a browser.

Use LibreOffice to create ODT, DOCX and PDFs

You already know that LibreOffice is the FOSS alternative to Microsoft Office. It has a word processor Writer (the Word alternative), spreadsheet application Calc (the Excel alternative), presentation slide maker Impress (the PowerPoint alternative), and a few other applications. While LibreOffice hoots and toots like a regular GUI application, it also has a demure command-line interface.

To convert the afore-mentioned HTML document to ODT format, type:

```
libreoffice --convert-to "odt" jokebook.html
```

You can use the same HTML document and convert it to DOCX so that Microsoft Office users can feel happy. (Microsoft Word can edit ODT files just fine.)

```
libreoffice --convert-to "docx:MS Word 2007 XML" jokebook.html
```

You can use the ODT or DOCX file that you generated and convert it to a PDF file:

```
libreoffice --convert-to "pdf" jokebook.odt
```

Why not convert to PDF straight from the HTML? Why create the intermediate ODT or DOCX file? Because the HTML document does not have any concept of page size, margins, headers, footers and all that jazz.

Create documents with images

When you convert a HTML document containing images, the resultant ODT or DOCX documents will show the images all right. When you move the documents or mail them to someone, the images will disappear. This is because the images in the ODT or DOCX documents continue to be loaded from the source image files. To fix this problem, you need to encode the images as text. This is similar to how images and attachments are encoded in email messages — using base64 encoding. View the message source of an email containing an attachment; you will find the file encoded as plain text.

Instead of using an image file like this:

```
<img src="lion-and-deer.png" />
```

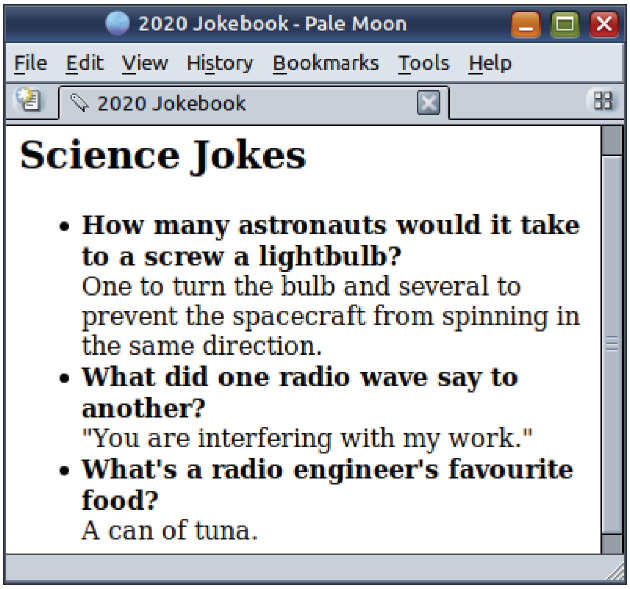

Figure 2: CommonMark generates HTML tags on a strictly as-needed basis. You have to use that generated content in a pre-existing page or HTML template

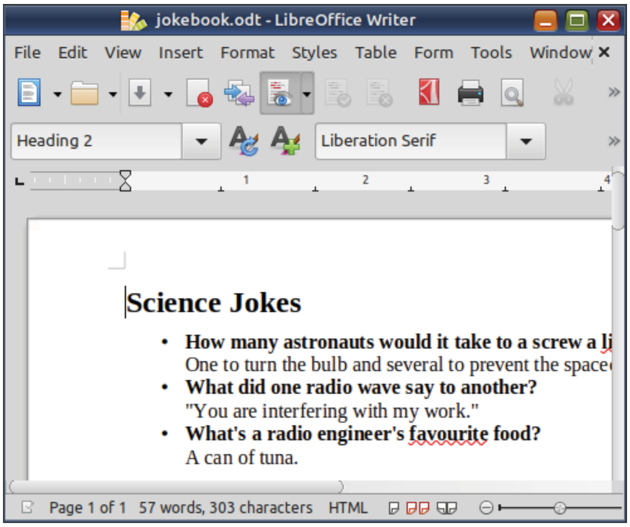

Figure 3: The command-line interface of LibreOffice can be used to automate the conversion of HTML documents to ODT, DOCX or PDF documents

… you can encode it as text like this …

```
<img src="data:image/png;base64,iVBORw0KGgoAAAA…" />
```

That is not all. I have truncated the actual text of the encoded image. The full text is nearly 600 lines. If you are curious, try a command like this:

```
base64 lion-and-deer.png
```

You do not have to dirty your hands with manual text-encoding. LibreOffice will encode the images as text.